

Prior-Context Prosody Probe

Does pre-pended paragraph context change target-sentence prosody?

Acoustic analysis of 3 TTS models × 4 conditions × 5 repetitions

— 60 target-sentence segments extracted via VAD boundary detection

1 Caveat

This report describes acoustic measurements from one fixed target sentence per condition. The sample size is small (n=5 per condition per model). All metrics are acoustic proxies (parselmouth F0, librosa pyin, VAD pauses). Informal listening was used only as a segmentation and repetition sanity check, not as a controlled listening study. Results are exploratory and hypothesis-generating.

2 Methodology

Design

4 conditions: cold (target sentence only), primed_neutral (target preceded by a neutral paragraph), primed_owl (target preceded by an owl-description paragraph), primed_death (target preceded by a death/funeral paragraph).

Target sentence (fixed across all conditions): *"The quarterly figures were reviewed and submitted before the deadline."* (10 words).

Prime paragraphs: 4 sentences, ~180-190 characters each. Neutral, owl, and death primes are matched for sentence count and punctuation pattern but differ in semantic valence.

Models

Chatterbox and **XTTS-v2** use VCTK p229 female reference for speaker identity with fixed seed-derived variation across repetitions. **Kokoro** uses af_bella voice preset with `split_pattern=None` to prevent newline-based segmentation.

Pipeline

Full clips generated (prime + target as one input) → target sentence extracted via VAD pause-boundary detection → 8 prosodic features measured → one-way ANOVA + pairwise t-tests with Bonferroni correction.

Acoustic Features (8)

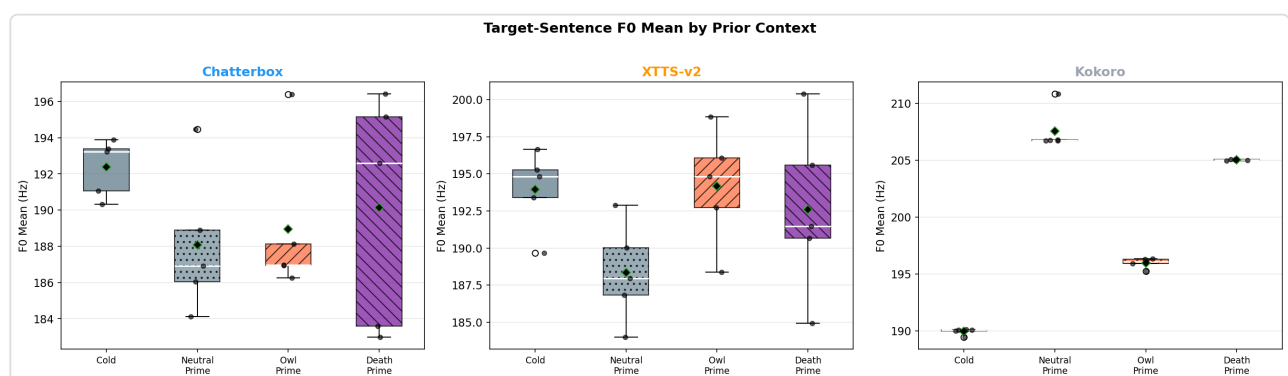
Feature	Tool	Captures
F0 mean	parselmouth (Praat)	Average pitch
F0 std	parselmouth	Pitch variance
F0 range	parselmouth	Pitch excursion
Speech rate	word count / voiced duration	Speaking pace
Pause count	RMS threshold (0.02)	Pause frequency
Pause duration	RMS threshold	Average pause length
RMS energy	librosa	Loudness
Spectral centroid	librosa	Voice brightness

Statistics

One-way ANOVA per model per feature (4 conditions). Post-hoc pairwise Welch t-tests with Bonferroni correction (6 comparisons per feature $\times$ 8 features = 48 tests per model). Effect sizes: η^2 for ANOVA, Cohen's d for pairwise differences. These tests are descriptive for this single-sentence probe, not confirmatory population-level inference.

3 F0 Mean: Primary Metric

Average pitch of the target sentence across conditions. The hypothesis predicts that semantically different primes produce different F0 in the identical target sentence.



ANOVA Summary

Model	F(3,16)	p	η^2	Sig?
Chatterbox	0.90		0.4637	0.144 No
XTTS-v2	2.19		0.1288	0.291 No
Kokoro	369.77	0.0000 ***	0.986	Yes

Kokoro: Pairwise Comparisons (Bonferroni-corrected)

Pair	Mean A	Mean B	t	p (corr)	Cohen's d
Cold vs Neutral Prime	190.0	207.6	-21.40	1.15e-04 ***	-13.54
Cold vs Owl Prime	190.0	196.0	-25.45	3.34e-07 ***	-16.10
Cold vs Death Prime	190.0	205.0	-116.46	6.58e-08 ***	-73.66
Neutral Prime vs Owl Prime	207.6	196.0	13.84	4.54e-04 ***	8.75
Neutral Prime vs Death Prime	207.6	205.0	3.13	0.2110	1.98
Owl Prime vs Death Prime	196.0	205.0	-44.91	6.36e-06 ***	-28.40

Kokoro shows a large target-sentence F0 shift under prior context. The cold condition (~190 Hz) differs from all primed conditions (~196-208 Hz, $p < 0.001$). Some pairwise differences among primed conditions also appear, but the dominant separation is cold vs anything-with-context. This supports a "has prior paragraph" context effect more strongly than a specific semantic priming effect.

Chatterbox and XTTS-v2: no significant F0 condition effect. Stochastic variation between repetitions exceeds any between-condition difference. With $n=5$ per condition, the experiment is underpowered to detect small effects in these models.

4 All Features: ANOVA Results

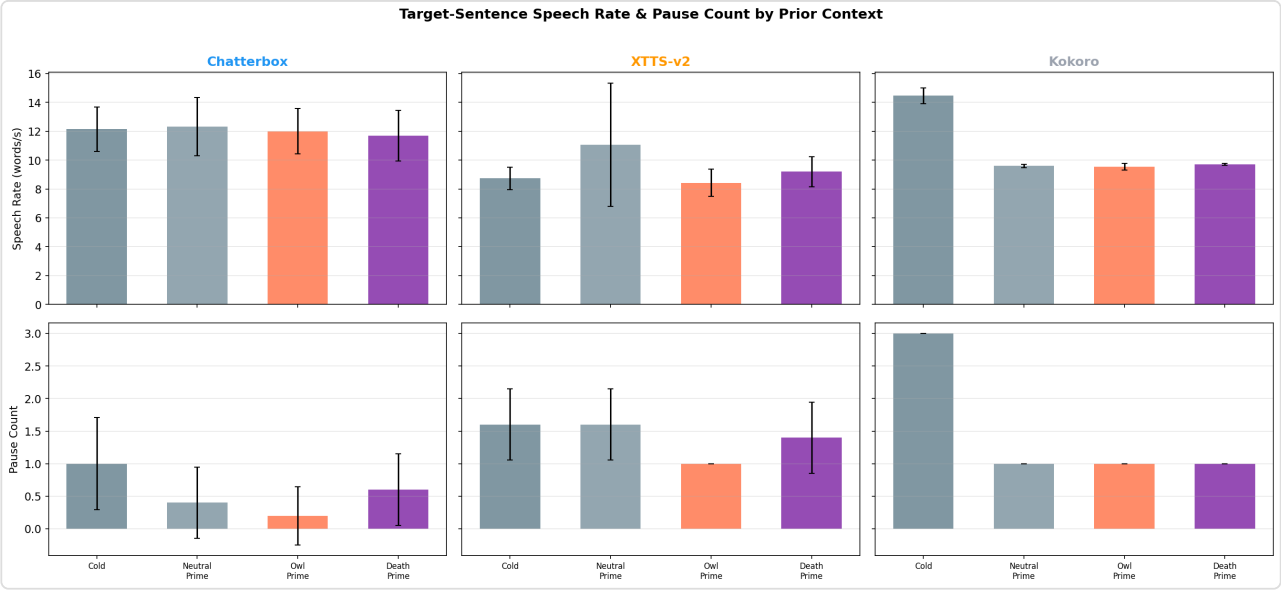
One-way ANOVA (4 conditions) for each feature across all 3 models. Significant effects ($\alpha=0.05$) are bolded.

Model	Feature	F	p	η^2	Sig
Chatterbox	F0 Mean	0.90	0.4637	0.144	No
Chatterbox	F0 Std	1.77	0.1940	0.249	No
Chatterbox	F0 Range	0.51	0.6800	0.088	No
Chatterbox	Speech Rate	0.12	0.9490	0.021	No
Chatterbox	Pause Count	1.79	0.1887	0.252	No
Chatterbox	Pause Duration	2.09	0.1423	0.281	No
Chatterbox	RMS Energy	0.23	0.8747	0.041	No
Chatterbox	Spectral Centroid	1.34	0.2970	0.201	No
XTTS-v2	F0 Mean	2.19	0.1288	0.291	No
XTTS-v2	F0 Std	0.34	0.7989	0.059	No
XTTS-v2	F0 Range	0.32	0.8085	0.057	No
XTTS-v2	Speech Rate	1.34	0.2961	0.201	No
XTTS-v2	Pause Count	1.78	0.1919	0.250	No
XTTS-v2	Pause Duration	0.73	0.5486	0.121	No
XTTS-v2	RMS Energy	3.22	0.0509	0.376	No
XTTS-v2	Spectral Centroid	0.44	0.7258	0.077	No
Kokoro	F0 Mean	369.77	0.0000 ***	0.986	Yes
Kokoro	F0 Std	20.88	8.87e-06 ***	0.797	Yes
Kokoro	F0 Range	1421.49	0.0000 ***	0.996	Yes
Kokoro	Speech Rate	310.95	0.0000 ***	0.983	Yes
Kokoro	Pause Count	inf	0.0000 ***	1.000	Yes
Kokoro	Pause Duration	3128.27	0.0000 ***	0.998	Yes
Kokoro	RMS Energy	118029.12	0.0000 ***	1.000	Yes
Kokoro	Spectral Centroid	12274.80	0.0000 ***	1.000	Yes

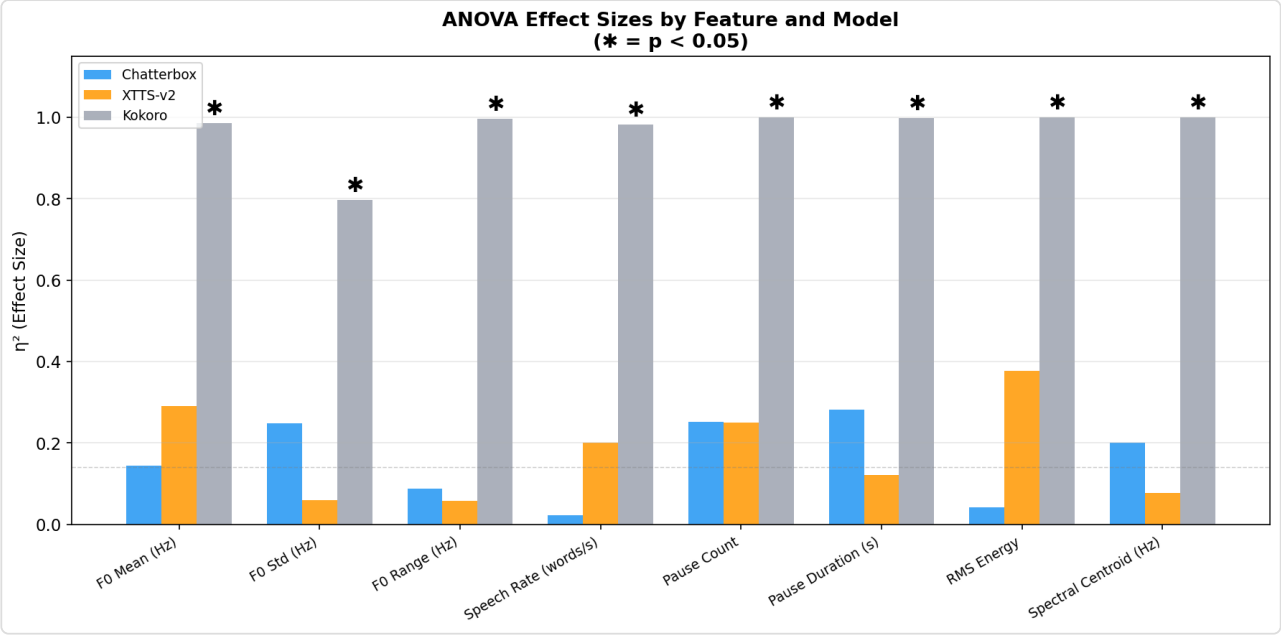
Significant features per model: Chatterbox = 0/8, XTTS-v2 = 0/8, Kokoro = 8/8. Kokoro shows strong across-the-board context effects. Chatterbox and XTTS-v2 show no reliable condition effects on any feature.

5 Visual Summary

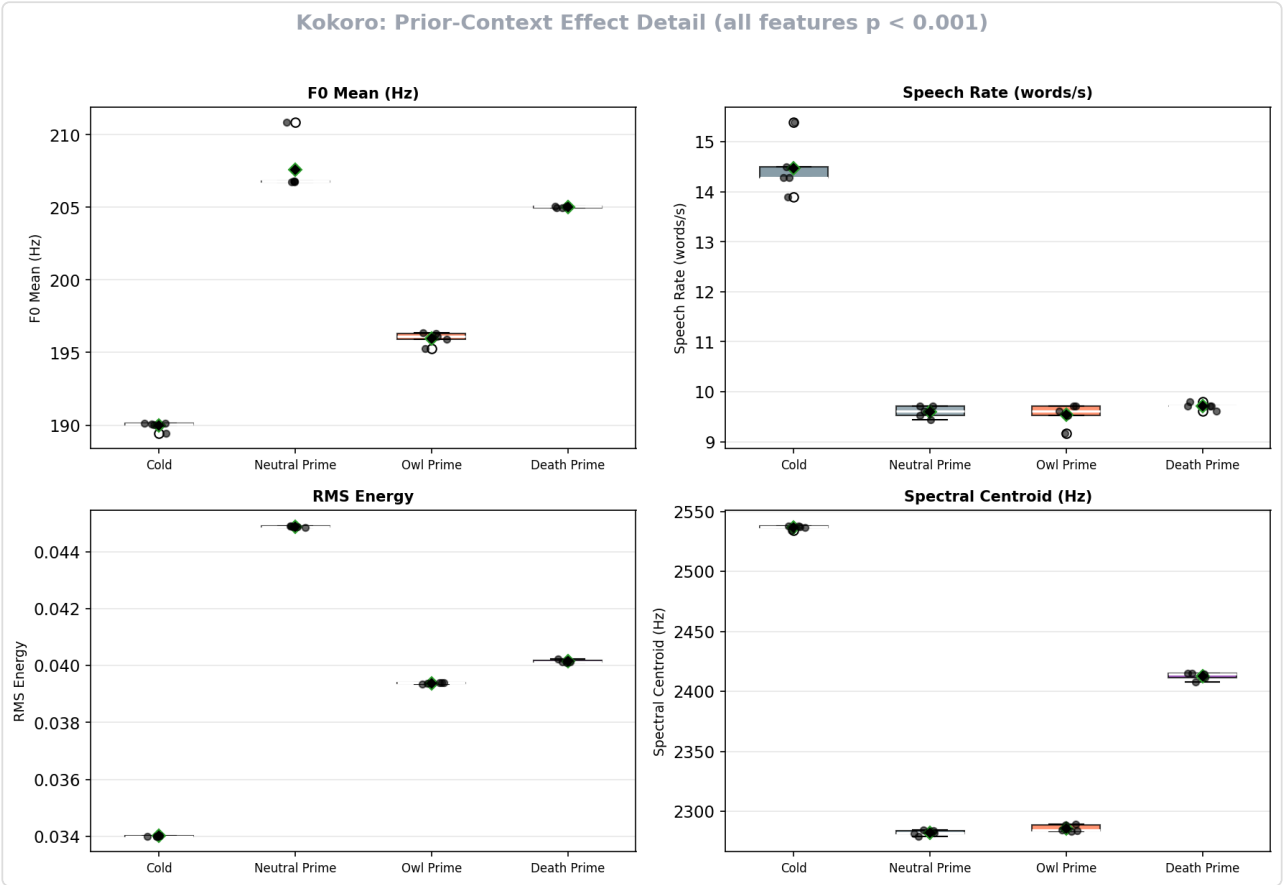
Speech Rate & Pause Count



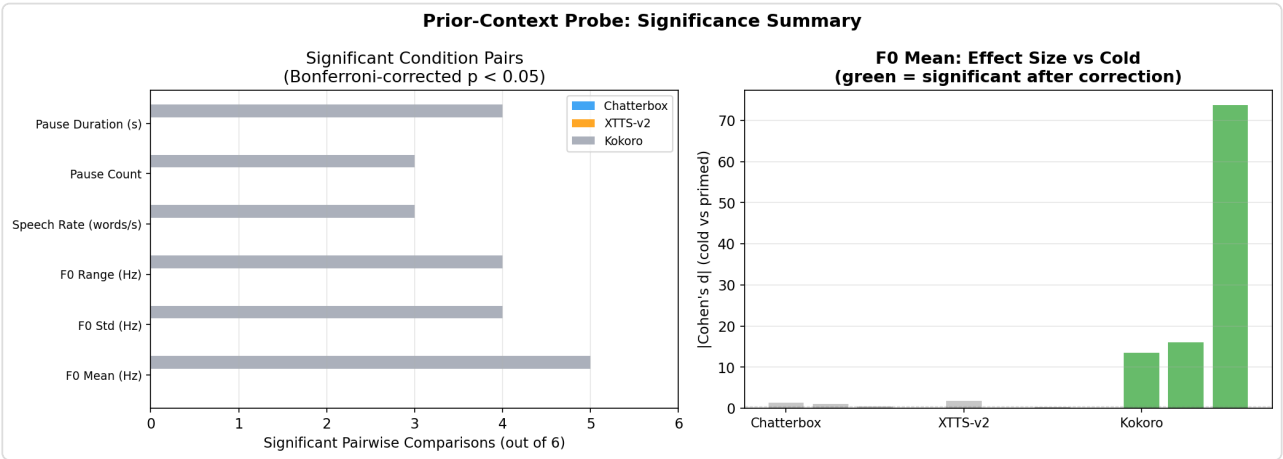
Effect Sizes (η^2) by Feature



Kokoro: Key Features Detail



Significance Summary



6 Condition Means

Mean values per condition per model for primary features. Standard deviations in parentheses (n=5).

Chatterbox

Feature	Cold	Neutral Prime	Owl Prime	Death Prime
F0 Mean (Hz)	192.4 (1.6)	188.1 (4.0)	188.9 (4.2)	190.1 (6.4)
Speech Rate (words/s)	12.2 (1.5)	12.3 (2.0)	12.0 (1.6)	11.7 (1.7)
RMS Energy ()	0.1 (0.0)	0.1 (0.0)	0.1 (0.0)	0.1 (0.0)
Spectral Centroid (Hz)	1363.5 (32.2)	1391.5 (35.9)	1338.1 (59.2)	1377.8 (44.3)

XTTS-v2

Feature	Cold	Neutral Prime	Owl Prime	Death Prime
F0 Mean (Hz)	194.0 (2.7)	188.3 (3.3)	194.2 (3.9)	192.6 (5.8)
Speech Rate (words/s)	8.7 (0.8)	11.1 (4.3)	8.4 (0.9)	9.2 (1.1)
RMS Energy ()	0.1 (0.0)	0.1 (0.0)	0.1 (0.0)	0.1 (0.0)
Spectral Centroid (Hz)	1521.8 (47.4)	1494.2 (71.1)	1531.4 (50.9)	1501.0 (61.8)

Kokoro

Feature	Cold	Neutral Prime	Owl Prime	Death Prime
F0 Mean (Hz)	190.0 (0.3)	207.6 (1.8)	196.0 (0.4)	205.0 (0.1)
Speech Rate (words/s)	14.5 (0.6)	9.6 (0.1)	9.5 (0.2)	9.7 (0.1)
RMS Energy ()	0.0 (1.38e-05)	0.0 (2.43e-05)	0.0 (1.78e-05)	0.0 (4.76e-05)
Spectral Centroid (Hz)	2536.8 (1.6)	2282.3 (2.1)	2286.0 (2.7)	2412.8 (3.1)

7 Methodological Limitations

- **Sample size.** n=5 per condition is minimal for statistical inference. Between-condition differences require large effects to reach significance.
- **Single sentence.** One fixed target sentence means results may not generalise. The sentence structure, word choice, and phonetic content may interact with priming sensitivity.
- **VAD segmentation.** Target-sentence extraction uses energy-based pause detection with a duration heuristic. No forced alignment validates that segments correspond exactly to the target sentence text.
- **Kokoro near-determinism.** Kokoro cold condition shows near-zero variance across repetitions (F0 mean SD = 0.28 Hz). This inflates statistical significance for Kokoro comparisons due to vanishing

within-group variance. Extremely large effect sizes partly reflect this unusually low within-condition variance and should not be read as population-level effect estimates.

- **Semantic vs syntactic priming.** The primed conditions differ in both semantic content and lexical features. The owl paragraph contains different phonemes, syllable counts, and word frequencies than the death paragraph. Observed differences may reflect acoustic properties of the prime text, not semantic transfer.
- **No controlled listening study.** Informal listening confirmed that target segments were natural and repetitions were not identical, but no blinded human rating or ABX test was run.
- **Multiple comparisons.** 24 ANOVAs and 144 pairwise tests with Bonferroni correction still carry elevated Type I error risk.
- **Kokoro voice mismatch.** Kokoro uses a different speaker (af_bella) than the VCTK p229 reference used by Chatterbox and XTTS-v2. Cross-model comparisons are confounded by speaker identity.

8 Conclusions

Kokoro shows strong context carryover (all 8 features significant). Cold F0 mean (190.0 Hz) is reliably lower than all primed conditions. F0 range increases $\times 1.3$, speech rate drops -34% , and pause patterns reorganise. This is a clear "has prior context" effect in this single-sentence probe.

However, the primed conditions themselves (neutral, owl, death) differ only subtly from each other in Kokoro. The separation is primarily cold vs everything-else, suggesting that the mere *presence* of preceding text, rather than its semantic content, drives the effect. Extremely large effect sizes partly reflect Kokoro's low within-condition variance.

Chatterbox (0/8) and XTTS-v2 (0/8): no reliable condition effect. Between-repetition stochastic variation exceeds any between-condition signal. These models' prosody for a fixed 10-word sentence showed within-condition variability larger than the observed between-condition differences. This does not mean priming is absent — it may exist below the detection threshold of $n=5$.

What This Tells Us

- Preceding paragraph context can measurably affect target-sentence prosody in at least one tested model: Kokoro.
- The strongest effect is a binary "has context / no context" signal, not a clean graded semantic response.
- Evidence for semantic *content*-specific priming (neutral vs owl vs death) is weak. The neutral-prime condition produces broadly similar prosody to the emotionally charged primes.
- The tested systems exhibited different levels of context carryover under these conditions. Architecture, voice preset, speaker reference, and pipeline differences are confounded.
- For this single target sentence, prior text mattered for Kokoro but not detectably for Chatterbox or XTTS-v2.

Practical Recommendation

If building a prosody-sensitive TTS pipeline, do not assume preceding text is acoustically neutral. Test target-only outputs after segmentation and include a neutral-prime control. For Chatterbox and XTTS-v2, any context signal in this experiment is smaller than within-condition stochastic variance at $n=5$.

Prior-Context Prosody Probe — 60 target-sentence segments across Chatterbox, XTTS-v2, and Kokoro.
Report generated with Python 3.12 • matplotlib • weasyprint.